



Ghinwa ISMAIL

PhD Researcher | Digital Twins for 5G/IoT | Machine Learning in Wireless Networks | SDN/Edge Systems Expert

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Strasbourg, France.

PhD candidate in Telecommunications at the University of Strasbourg, conducting research at ICube Lab on the integration of Digital Twins and Machine Learning into next-generation wireless networks (5G, IoT, IIoT). My work explores how digital replicas of network environments can enhance performance, reliability, and self-optimization in real-time wireless systems. With a multidisciplinary background spanning data science, SDN/NFV, and embedded systems.

EDUCATION

PhD in Telecommunications

Since November 2024

University of Strasbourg, ICube Lab, Strasbourg, France

Wireless network modeling, self-calibrating digital twins, SDN-based orchestration, energy-efficient monitoring, integration with IETF RAW initiatives.

Master's Degree in Data Science and Network Intelligence, Double Degree Program.

From September 2023 to August 2024

Institut Polytechnique "Télécom SudParis", Évry, France

Blockchain for Data Scientists, Business Modeling and ICT, Data Science "Theory to practice", Data Visualization, Deep Learning, Transfer Learning - application to NLP and Image Recognition, Internet of Things and Digital Twins, Network Intelligence and Communication Services, Service-Oriented Computing, Wireless Access Networks: State of the Art, Challenges and Evolutions

Master's Degree in Telecommunication Engineering.

From September 2022 to October 2024

Universita della Calabria, Rende, CS, Italy

IoT Device Programming, Wireless Device and Network, Antennas and Propagation, Simulation and Performance Optimization, IoT System for Environmental Monitoring, Measurement for Telecommunication Systems, Networking Aspects of Internet of Things, Distributed Systems and Cloud/Edge Computing.

Bachelor's Degree in Telecommunication Technologies.

From September 2016 to December 2020

Tishreen University, Latakia, LA, Syria

Achieved a remarkable final grade of 90.5%, with a thesis project titled "Heart Patients Device."

WORK EXPERIENCE

Internship

From February 2024 to August 2024

Orange Innovation, Caen, France

During my internship at Orange Innovation, I focused on researching distributed communication systems to advance current methodologies. I improved the integration and functionality of Smart Contracts on the Ethereum blockchain through API optimization. Additionally, I managed CI/CD pipelines with GitLab and Kubernetes, ensuring efficient and reliable software deployment. My work culminated in the development of a functional prototype that validated the research theories, showcasing potential enhancements in communication technologies. This role deepened my understanding of blockchain applications and agile development in the tech research domain.

WORK EXPERIENCE

Teacher Assistant

From 2020 to 2022

Tishreen University, Latakia, Syria

- Demonstrated expertise in Electronic Measurements, Network Management, Security, and Computer Skills to third-year and first-year students.
- Implemented cutting-edge simulation programs like OptiSystem and VMware for effective concept illustration.

Mathematics Teacher

From 2016 to 2020

Private Jop, Latakia, Syria

- Facilitated students' understanding of mathematics, contributing to their academic growth.

PROJECTS:

Natural Language Processing, Telecom Sudparis2024:

SHROOM (SemEval-2024 Task-6) - Competition Participant (2024)

Gained experience in identifying and classifying inaccurate outputs from Natural Language Generation models and enhanced my understanding of the challenges and limitations of NLG systems.

Predictive Analysis of Video Streaming Quality and User Satisfaction Telecom Sudparis 2023:

The Project analyzed a video streaming dataset to identify key factors

influencing user satisfaction, measured by the Mean Opinion Score (MOS).

After preprocessing the data, I used three predictive models (Random Forest, XGBoost, and SVC).

Advanced Sustainable Agriculture Initiative, Universita Della Calabria 2023:

Led the development of a sustainable agriculture initiative, incorporating automated systems to enhance temperature control, lighting management, and irrigation optimization within a greenhouse environment.

Designing Array Antenna, Universita Della Calabria 2023:

This project involves the design and optimization of a four-element array antenna for the Antennas and Propagation course. Key objectives include achieving a -22 dB S11 return loss, a 0.4 GHz bandwidth at -10 dB, operating at 10 GHz, and directing the radiation pattern at -5 degrees. Successful completion demonstrates expertise in antenna design principles and practical application.

Android Application Programming, Universita Della Calabria 2023:

This project entails the development of an online shop application. It aims to provide a user-friendly platform for customers to browse, select, and purchase products or services. The application will incorporate features such as product catalog, shopping cart, secure payment processing, and order management.

Continuous Cardiac Monitoring and Emergency Response System, Tishreen University 2019:

Spearheaded the development of an innovative device designed to provide continuous heart rate monitoring and streamline emergency response coordination for patients with cardiac conditions.

SKILLS SUMMERY

Skills:

Data Analysis, Network Design and Management, Optic Fiber System Designing, Deep Learning, Internet of Things (IoT), Cloud Computing.

Programming Languages:

Python, C++, Matlab.

Protocols and Standards:

MQTT, Dijkstra, SIP, TCP/IP, VoIP.

Tools:

Ansoft Designer, Packet Tracker, Arduino IDE, Android Studio.

SKILLS SUMMARY

Platforms:

Nod-Red, Web, Windows, Cloud.

Soft Skills:

Leadership, Attention to Detail, Creativity, Teamwork and Collaboration, Time Management.

HONORS & AWARDS

- AL-Basil Certificate for the desolate graduate with Excellent appreciation (2020).
- AL-Basil Certificates with the first grade (2017, 2018, 2019).
- Recognized for participation in the AI-Basel Fair for invention and innovation (2021).
- Honored for contributing to the third applied college exhibition (2019).

RESEARCH EXPERIENCE

Internship at Orange Innovation

Location: Caen, France, Since February 2024.

Role: Research Intern

Description: Engaged in cutting-edge research on distributed communication solutions, focusing on the application of Smart Contracts on the Ethereum blockchain. Developed a functional prototype that leverages blockchain technology to enhance security and efficiency in distributed communications.

Key Contributions: Designed and implemented a novel API for Smart Contracts, significantly improving transaction security and efficiency. This work represents a substantial advancement in blockchain applications for distributed communications, showcasing my ability to innovate and apply data science principles in real-world scenarios.

Bed Bugs Infestation: Monitoring and Collaborative Solutions

Institution: Telecom SudParis, 2023.

Role: Project Member.

Description: Spearheaded a pioneering project aimed at revolutionizing pest management in the hospitality industry. Developed an advanced bed bug detection system integrating YOLO for real-time image processing, Federated Learning for data privacy, and Random Forest algorithms for issue prioritization.

Innovative Contributions:

- Implemented YOLO algorithms for accurate, real-time detection of bed bugs, significantly enhancing pest detection capabilities.
- Utilized Federated Learning to ensure data privacy and security, enabling on-site AI model training without compromising sensitive information.
- Employed Random Forest algorithms for efficient classification and prioritization of pest issues, facilitating prompt and targeted pest control interventions.

LANGUAGES

Arabic

Native language

English

Advanced

French

beginner

Italian

Elementary